

Basic Commands

Prerequisite
Install Linux kernel header package(s) if you want to collect data using the LiKI DKLM module. See the LinuxKI MasterClass documentation for more details.

Installation:
```
rpm --install --nodeps linuxki-8.0-1.noarch.rpm
dpkg --install linuxki_8.0-1_all.deb
```

Add LinuxKI to your PATH variable:
```
export PATH=$PATH:/opt/linuxki
```

Collect KI Dump:
```
cd /dev/shm                  # if memory is available
runki [-d <secs>]            # default is 20 secs
```

IMPORTANT NOTE: The longer the trace collection, the larger the resulting LinuxKI trace dump. Be sure you have plenty of disk space or space in /dev/shm to collect your LinuxKI trace dump!

Post-process KI Dump data
```
kiall [-r]
```

Mounting the debug filesystem (debugfs)
```
mount -t debugfs debugfs /sys/kernel/debug
```

Note: Most Linux versions now mount /sys/kernel/debug by default.

Curses-base live mode
```
kiinfo -live [-a <secs>]     # default is 5 secs
```

Clean-up KI files in cwd
```
kiclean -r -p -f
```

KI dump collection Examples

KI dump help
```
runki -h
```

Collect for 60 seconds
```
runki -d 60
```

Collect using ftrace with all events
```
runki -f
```

Collect with Advanced CPU statistics
```
runki -R
```

Collect and include scsi subsys
```
runki -e default -s scsi
```

Collect with Collect/MW, sar, perf data and Java stacks
```
runki -M -U -X -j
```

Collect with all subsystems, excluding some system calls
```
echo "time" >sysignore
echo "futex" >>sysignore
echo "gettimeofday" >>sysignore
runki -e all -I sysignore
```

Collect using PID filtering for 2 minutes
```
runki -P 26814 -d 120
```

Collect data and add a comment
```
runki -c "slow run, only 20MB/sec"
```

Continuous Logging, then dump on likiend
```
runki -d 0
kiinfo -likiend
```

KI Dump Post Processing Examples

Post-processing help
```
kiall -h
```

Create <hostname>/<timestamp> subdir structure
```
kiall -r
```

Create Kparse Report in text format
```
kiall -x
```

Online kiinfo Examples

kiinfo help
```
kiinfo -h
```

PID Analysis Report with multiple PID filters for 30 seconds
```
kiinfo -kipid pid=6814,pid=6815,scdetail -a 30
```

Disk Analysis report to show top 10 pids accessing a specific dev every 5 seconds for a minute
```
kiinfo -kidsk dev=0x00800010,npid=10 -a 5 -p 12
```

CPU/RunQ Analysis with Advanced CPU statistics
```
kiinfo -kirunq events=default,msr -a 5
```

Show just SCSI Synchronize Cache commands for 1 minute
```
kiinfo -kitrace subsys=scsi -a 60 | grep -i synchronize_cache
```

KI Dump kiinfo Examples

PID Analysis Report with System Call details, Cooperating Task details, and per-pid RunQ Histogram
```
kiinfo -kipid scdetail,coop,rqhist -ts 0416_1523
```

PID Analysis Report with Time filtering
```
kiinfo -kipid scdetail -start 3.0 -end 4.0 -ts 0416_1523
```

Disk Analysis Report with modified I/O Histogram
```
echo "2 5 10 20 50 100" >bkfname
kiinfo -kidsk bkfname=bkfname -ts 0416_1523
```

Kitrace Report with formatted time and the per-cpu sequence counter for a specific CPU
```
kiinfo -kitrace seqcnt,fmttime,cpu=4 -ts 0416_1523
```

Kiinfo -live Commands Menu

Global Statistics and Select

```
s - Select Task/CPU/Disk
g - Global Task List
l - Global Node Stats
c - Global CPU Stats
p - Global Prof Stats
h - Global HT CPU Stats
i - Global IRQ Stats
d - Global Disk Stats
m - Global Mpath Stats
y - Global WWN Stats
z - Global HBA Stats
t - Global IO by PID
f - Global File Stats
w - Global Wait Stats
u - Global Futex Stats
n - Global Socket Stats
k - Global Container Stats
```

Per-Task Statistics and Selective Statistics

```
G - Task Main Stats
D - Task Disk Stats
M - Task Mpath Stats
L - Task System Calls
W - Task Wait Stats
P - Task Profile Stats
F - Task File Stats
O - Task Coop Stats
U - Task Futex Stats

C - Select CPU Stats
T - Select Disk Stats
I - Select IRQ Stats
K - Select Container Stats
X - Select Futex Stats
```

Control and Miscellaneous Commands

```
? - Help
r - Refresh
b - Prev Screen
+ - Show Syscall Detail
- - Hide Syscall Detail
```

Online mode only

```
a - Set Alarm Interval
e - Exclude Syscall
E - Show Excl Syscalls
```

Dump mode only from LiKI traces

```
> - Next Step
< - Prev Step
j - Step Time
J - Jump to Time
```

runki [-h] [-L] [-M] [-U] [-X] [-a] [-j] [-J path] [-f | -p] [-n interface] [-v] [-d duration] [-t maxrun] [-P pid] [-G tgid] [-C cpu] [-D dev] [-R] [-e event] [-s subsys] [-I sysignore] [-T timestamp] [-c "comment"]

```
-h              Help
-L              Gather minimal data for local analysis...no gzip/tar
-Z              Use parallel gzip to speed up gzip/tar phase (NOT recommended on production systems)
-M              Include Collectl/MeasureWare data collection
-U              Include userspace profile (perf) data collection
-X              Include sar data collection
-a              Execute 'perf annotate' on userspace profile (perf) data
-j              Collect Java stacks
-J <path>       Location of Java jstack command
-n <interface>  Gather tcpdump trace data
-v              Skip vxfs, misc disc data collection
-d <secs>       Duration of KI data collection. Defaults to 20 secs if not specified.
-d 0            Initiate continuous logging.  Collect LinuxKI dump with kiinfo -likiend
-t <secs>       Maximum time for data collection tools to run. Defaults to 120 secs if not specified.
-p              Skip Per-PID data (lsof, stacks, numa_maps, maps)
-f              Use ftrace tracing instead of liKI DLKM tracing to dump trace data.
-P <pid>        Filter collection on Task ID or PID (LiKI only)
-G <tgid>       Filter collection on Task Group ID or TGID (LiKI only)
-C <cpu>        Filter collection on cpu (LiKI only)
-D <dev>        Filter collection on device (LiKI only)
-R              Advance CPU stats (Freq, CPI, LLC Hit%) using MSR registers (LiKI only)
-e <event>      Identify events to dump. Possible values are: default | all | <event>
-s <subsys>     Identify subsys to dump. For example: irq, scsi, block, etc.
-I <sysignore>  File to specify ignored system call (liki only)
-F <hc_per_sec> Number of CPU Profile events per second (default: 100)
-T <timestamp>  Should be of the form MMDD_HHMM
-V "<varargs>"  Specify multiple filters and flags to pass to kiinfo -likidump
-c "comment"    echo comment into comment.$tag file
```

kiall [-h] [-l] [-m] [-r] [-f] [-c] [-x] [-M] [-B] [-t timestamp]

```
-h              Help
-l              Lite version - bypasses some options
-m              Do not process collectl or MWA data
-r              restore into created subdir - <host>/<MMDD_HHMM>
-f              Find KI files in pwd and any directories below
-c              Cluster-wide processing
-x              Generate Kparse text format instead of html format
-Z              Use parallel gzip/gunzip to speed up gzip/gunzip phase
-B              Add Disk Block Frequency stats to Kparse report
-t <timestamp>  Only process KI budle with matching timestamp
```

kiclean [-h] [-r] [-f] [-p]

```
-h              Help
-r              Recursively traverse sudirs looking for KI data to archive
-Z              Use parallel gzip/gunzip to speed up gzip/gunzip phase
-p              Remove PIDS subdirectory
-f              Force remove misc/tmp files
```